devx labs · HACKaMINeD 2026

Problem Statement

The Skills Mirage

Build India's first open workforce intelligence system

From market signals to reskilling paths - India's 50 crore workers need this.

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Two Sides of a Crisis. Zero Connection.

India posts 1 crore job listings/month. 45%+ employers still can't find the right skills.

DEMAND SIDE — Enterprises

- Companies hire GenAI engineers, fire call-centre agents
- Signal exists in job postings - nobody reads it at sector scale for India
- Hiring patterns reveal strategy 6 months before press releases
- No open India-specific market intelligence exists

VS

SUPPLY SIDE - Workers

- 38-yr BPO lead in Pune gets a generic 'learn Python' suggestion
- Job title tells nothing about actual skills or aspirations
- Most tools ignore geography and local job supply entirely
- 8%+ graduate unemployment even in high-hiring cities

SCALE OF THE PROBLEM

The Numbers Behind the Crisis

0.5 Cr+

job listings posted in India every month

45%+

employers cannot find candidates with right skills

8%+

graduate unemployment in high-hiring cities

30 Cr

Indian workers in formal + informal workforce

₹0

cost to build - all data sources are public & free

48 hrs

to build something that outlasts this hackathon

SYSTEM ARCHITECTURE

Two Layers. One Live System.

Layer 1 feeds real-time signals into Layer 2. A change in one must propagate to the other automatically.

LAYER 1 - Job Market Dashboard

- Live scraping + Open source data: Naukri + LinkedIn India
- Tab A: Hiring trends by city/sector
- Tab B: Skills demand + gap analysis
- Tab C: AI Vulnerability Index (0–100)
- Displacement early warning signals

→ feeds

LAYER 2 - Worker Intelligence Engine

- Worker profile: title + city + XP + write-up
- Personal AI Risk Score (0–100)
- Week-by-week reskilling path
- AI Chatbot: English + Hindi
- Free resources: SWAYAM, NPTEL + global

Three Tabs. All Live. All Data-Derived.

Real scraping from Naukri + LinkedIn. Must refresh live during demo. No static CSVs.

Tab A : Hiring Trends

Volume trends by job category, city, sector.

Time range: 7d / 30d / 90d / 1yr.

Min 20 cities - Tier 2 + Tier 3 required.

Tab B : Skills Intelligence

Top 20 rising + declining skills (week-over-week).

Skill gap map: what's being hired vs what PMKVY/SWAYAM, trains for.

Tab C : AI Vulnerability Index

- Score every job 0–100
- Signals: hiring decline + AI tool
- mentions in JDs + role replacement ratio
- Heatmap view by city
- Trend: rising or falling risk?
- Methodology panel - visible,
- no black boxes

Example: BPO Voice — Pune: 87/100 Critical · Data Entry -Jaipur: 74/100 High · Data Analyst (AI Tools): 18/100 Low

What the Worker Tells You

No CV. No long forms. Four inputs - and the system does the heavy lifting.

1. Current Job Title

E.g. Senior Executive, BPO
Free text - NLP normalises it

2. City

Tier 2 + Tier 3 fully supported
Min 20 cities required

3. Years of Experience

Integer or short range
Weighted in risk scoring

4. Short Write-Up (100–200 words) - The Most Important Input

What they do day-to-day. What they are good at. What work they want to move toward.
Why it matters: Two 'Senior Executives' can have completely different skills. The write-up extracts implicit skills, tools, soft skills, and aspirations no job title reveals. Your NLP must demonstrably use this input - if the write-up is ignored, it counts as not implemented.

Risk Score → Reskilling Path → Chatbot

Three connected outputs from one worker profile.

AI Risk Score

74 / 100

HIGH RISK

- ↑ +8 vs 30 days ago
- BPO voice hiring -34% in Pune
- AI tool mentions in JDs +40%
- vs peers: top 15% at-risk

Reskilling Path

→ AI Content Reviewer
(Hiring in Pune - L1 verified)

- Wk 1–3: NPTEL Data Basics
- (IIT Madras, free, 6 hrs/wk)
- Wk 4–5: SWAYAM AI Fundamentals
- Wk 6–8: PMKVY Digital Mktg
- (Nagpur centre, Wardha Rd)
- Total: 8 weeks @ 10 hrs/wk

AI Chatbot (EN + HI)

Context-aware, not canned.
Must handle 5 question types:

- Why is my risk score high?
- Show paths under 4 months
- How many BPO jobs in Indore?
- Is this NPTEL cert recognised?
- मुझे कहाँ से शुरू करना चाहिए?

Sample Datasets - All open source

Source	What It Contains	Where to Get It
PLFS Microdata	~4 lakh individual records/yr - employment, sector, 2017–2024	microdata.gov.in
PMKVY Training Data	State/district trained, certified, placed - 2015–2024	data.gov.in → search 'PMKVY'
Naukri (Kaggle)	~5 lakh records: job title, skills, city, salary (2019)	kaggle.com → 'naukri job postings india'
Naukri Live Scrape	Real-time listings - Apify actor or Python BeautifulSoup	apify.com → Naukri scrapers
LinkedIn India Jobs	Company, role, seniority, skills - Apify LinkedIn scraper	apify.com → LinkedIn Jobs
NPTEL Catalog	2,400+ courses - subject, IIT institution, duration	nptel.ac.in (scrapeable)
SWAYAM Courses	2,000+ free courses, 56M+ enrollments	swayam.gov.in (scrapeable)
WEF Future of Jobs	India displacement forecasts by role category	weforum.org — free PDF

The Chatbot - 5 Question Types Required

will test this live. It must use the worker's actual profile + Layer 1 data.

01 ***"Why is my risk score so high?"***

Must cite specific Layer 1 signals - hiring decline %, AI tool mention rate - from the worker's city and role.

02 ***"What jobs are safer for someone like me?"***

Must query Layer 1 Vulnerability Index and return low-score roles that are actively hiring in the worker's city.

03 ***"Show me paths that take less than 3 months"***

Must regenerate paths with a time constraint, still validating target roles against live Layer 1 data.

04 ***"How many BPO jobs are in Indore right now?"***

Must query live Layer 1 data. Must return a real number from the pipeline, not a hallucinated answer.

05 ***"मुझे क्या करना चाहिए?" (Hindi)***

Full Hindi support — input in Hindi, response in Hindi, context-aware to the worker's profile and risk score.

WHAT WINNING LOOKS LIKE

A team that builds a scraper + chatbot without the intelligence layer will score in the bottom third.

1. Live dashboard with real signal

Refresh during demo. Show a specific data-backed finding: 'BPO voice in Pune fell 22% in 30 days, AI call-handling +40%.' Methodology visible on screen.

2. Risk score that reacts to data

Enter a profile → score in seconds. Change a Layer 1 parameter → score updates live. Pre-computed scores that never change will not pass.

3. A reskilling path that is specific

Real course links. Week-by-week schedule. Target role verified as actively hiring in that city from L1 data. 'Learn Python' is not a path.

4. Chatbot that understands context

Judges ask 3 live questions using the worker's actual profile + L1 data. Show at least one exchange in Hindi. Generic LLM responses = zero score.

Bonus

Functional bonus features earn extra Demo Conviction points.

Displacement Early Warning

A watchlist in Layer 1 that flags job categories where hiring is declining fast enough to warrant proactive outreach — before layoffs begin. Workers in at-risk roles get notified to start reskilling now, not after the fact.

Employer-Side View

Flip the tool: show companies the supply/demand gap — what skills exist in a city vs. what that sector is actually hiring for. Not just 'help workers' but 'show employers where the talent pipeline is breaking'.

Bonus features must be functional

SCORING RUBRIC

How You'll Be Judged

25%	**Technical complexity & architecture**	All what tech and AI integrations is carried?
20%	**Problem depth & real-world applicability**	How much depth is thought and implemented to make it super relatable
20%	**Reskilling Path Quality**	City-matched, week-by-week, real resources? Write-up demonstrably influences the path?
15%	**Chatbot Intelligence**	5 question types?Context-aware? Works in Hindi w/o switching to Eng?
10%	**Layer Integration**	L1 changes propagate to L2 live? Feedback loop visible during demo?
10%	**Layer 1 Live Signal Quality**	Running on real data? Vulnerability Index data-derived with visible methodology?

48 hours.

Build something India's 50 crore workers actually need.

- Layer 1: Live dashboard — Naukri + LinkedIn scrape → Hiring Trends + Skills Intel + AI Vulnerability Index (0–100)
- Layer 2: Worker profile (title + city + XP + write-up) → Personal Risk Score + Week-by-week Reskilling Path
- AI Chatbot: context-aware, 5 question types, works in English and Hindi
- Both layers must talk to each other — live, during the demo. No static pipelines.

Build the real thing. Good luck.

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